

Envelope:

A curve which touches each member of a given family of curves is called envelope of that family.

Procedures to find envelope for a given family of curves:

Case-1: Envelope of one parameter

Let us consider $y = f(x)$ to be the given family of curves.

Step #1: Differentiate w.r.t to the parameter partially and find the value of the parameter.

Step #2: By substituting the value of parameter in the given family of curves, we get the required envelope.

Special case: If the given family of curve is quadratic in terms of the parameter, then envelope is given by discriminant = 0.

Case-2: Envelope of two parameters

Let us consider $y=f(x)$ to be the given family of curves, and a relation connecting these two parameters.

Step #1: Obtain one parameter in terms of the other parameter from the given relation.

Step #2: Substitute in the given eqnⁿ of curve, so that the problem of two parameters converts to the problem of one parameter.

Step #3: Use one-parameter technique to obtain envelope for the given family of curves.

Problem:

Find the envelope of the family of straight line $y = mx + \sqrt{a^2 m^2 + b^2}$, m is the parameter.

Solⁿ:

Given eqnⁿ of the family of curves is

$$y = mx + \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow (y - mx)^2 = (\sqrt{a^2 m^2 + b^2})^2$$

$$\Rightarrow y^2 - 2mxy + m^2x^2 = a^2m^2 + b^2$$

$$\Rightarrow m^2(x^2 - a^2) - 2mxy + y^2 - b^2 = 0 \quad \text{--- (1)}$$

Step #1: Differentiating (1) partially w.r.t to the parameter m , we obtain

$$2m(x^2 - a^2) - 2xy = 0$$

$$\Rightarrow m = \frac{xy}{x^2 - a^2} \quad \text{--- (2)}$$

Step #2:

Substitute the value of m in the given family of curves (1)

$$\left(\frac{xy}{x^2 - a^2}\right)^2(x^2 - a^2) - 2\left(\frac{xy}{x^2 - a^2}\right)xy + y^2 - b^2 = 0$$

$$\Rightarrow \frac{x^2y^2}{x^2 - a^2} - \frac{2x^2y^2}{x^2 - a^2} + (y^2 - b^2) = 0$$

$$\Rightarrow \frac{x^2y^2 - 2x^2y^2}{x^2 - a^2} + (y^2 - b^2) = 0$$

$$\Rightarrow \frac{-x^2y^2}{x^2 - a^2} = -(y^2 - b^2)$$

$$\Rightarrow x^2 y^2 = (x^2 - a^2)(y^2 - b^2)$$

$$\Rightarrow x^2 y^2 = x^2 y^2 - a^2 y^2 - b^2 x^2 + a^2 b^2$$

$$\Rightarrow b^2 x^2 + a^2 y^2 = a^2 b^2$$

$$\Rightarrow \frac{b^2 x^2}{a^2 b^2} + \frac{a^2 y^2}{a^2 b^2} = \frac{a^2 b^2}{a^2 b^2}$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The envelope of the given family of straight lines is an ellipse.

Ans)

Problem:

Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$, where a and b are two parameters which are connected by the relation $a+b=c$.

Soln:

Given eqn of family of straight lines is

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{--- (1)}$$

Also, $a + b = c$

$$\Rightarrow b = c - a \quad \text{--- (2)}$$

Substituting (2) into (1) to obtain

$$\frac{x}{a} + \frac{y}{c-a} = 1 \quad \text{--- (3)}$$

Step # 1: Differentiating (3) partially w.r.t to the parameter a , we obtain

$$-\frac{x}{a^2} + \frac{y}{(c-a)^2} = 0$$

$$\Rightarrow \frac{x}{a^2} = \frac{y}{(c-a)^2}$$

$$\Rightarrow \frac{(c-a)^2}{a^2} = \frac{y}{x}$$

$$\Rightarrow \frac{c-a}{a} = \sqrt{\frac{y}{x}}$$

$$\Rightarrow \frac{c}{a} - 1 = \sqrt{\frac{y}{x}}$$

$$\Rightarrow \frac{c}{a} = 1 + \sqrt{\frac{y}{x}} = \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x}}$$

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$$\Rightarrow \frac{a}{c} = \frac{\sqrt{x}}{\sqrt{x} + \sqrt{y}}$$

$$\Rightarrow a = \frac{c\sqrt{x}}{\sqrt{x} + \sqrt{y}}$$

Now, substituting the value of a into (2), we have

$$b = c - \frac{c\sqrt{x}}{\sqrt{x} + \sqrt{y}}$$

$$\begin{aligned} \Rightarrow b &= c \left(1 - \frac{\sqrt{x}}{\sqrt{x} + \sqrt{y}} \right) \\ &= c \left(\frac{\cancel{\sqrt{x}} + \sqrt{y} - \cancel{\sqrt{x}}}{\sqrt{x} + \sqrt{y}} \right) \end{aligned}$$

$$\Rightarrow b = \frac{c\sqrt{y}}{\sqrt{x} + \sqrt{y}}$$

Step # 2: Substituting the value of a & b into

① we obtain

$$\frac{x}{\frac{c\sqrt{x}}{\sqrt{x} + \sqrt{y}}} + \frac{y}{\frac{c\sqrt{y}}{\sqrt{x} + \sqrt{y}}} = 1$$

$$\Rightarrow \frac{x(\sqrt{x} + \sqrt{y})}{c\sqrt{x}} + \frac{y(\sqrt{x} + \sqrt{y})}{c\sqrt{y}} = 1$$

$$\Rightarrow \sqrt{x}(\sqrt{x} + \sqrt{y}) + \sqrt{y}(\sqrt{x} + \sqrt{y}) = c$$

$$\Rightarrow (\sqrt{x} + \sqrt{y})(\sqrt{x} + \sqrt{y}) = c$$

$$\Rightarrow (\sqrt{x} + \sqrt{y})^2 = c$$

$$\Rightarrow \sqrt{x} + \sqrt{y} = \sqrt{c}$$

which is the required envelope.

Ans

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